

Generation of artificial earthquake records with a nonstationary Kanai–Tajimi model

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Abstract

The generalized nonstationary Kanai–Tajimi model is used to describe and simulate the ground motion time histories. Both amplitude and frequency nonstationarities are incorporated in the model. The moving time-window technique is used to evaluate the time varying parameters of the model using actual earthquake records. Application of the model for several Iranian earthquakes Naghan (1977), Tabas (1978) and Manjil (1990) are presented. It is shown that the model and identification algorithms are able to accurately capture the nonstationary features of these earthquake accelerograms. The statistical characteristics of the spectral response curves of the simulated accelerograms are compared with those for the actual records to demonstrate the effectiveness of the model. © 2001 Elsevier Science Ltd. All rights reserved.

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1. Introduction

For earthquake resistant design of critical structures, a dynamic analysis, either response spectrum or time history is frequently required. The major drawback of the response spectrum analysis in seismic design of structures lies in its inability to provide temporal information of the structural responses. Such information is sometimes necessary in achieving a satisfactory design. On the other hand, different building codes, require a dynamic analysis in certain cases such as existence of irregular features in building plan, non-uniform spatial distribution of mass or stiffness over the height of the buildings, etc. Also, in the design of critical facilities or major structures such as nuclear power plants, dams, or even high-rise buildings, the final design is usually based on a complete linear or nonlinear time-history analysis.

To provide input excitations to structural models for sites with no strong ground motion data, it is necessary to generate artificial accelerograms. It has long been

established that due to parameters such as geological conditions of the site, distance from the source, fault mechanism, etc. different earthquake records show different characteristics. Thus, the simulated earthquake records must have realistic duration, frequency content, and intensity, representing the physical conditions of the site. Therefore, it is reasonable to identify certain parameters to measure the resulting differences in the strong ground motion data. This will make it possible to generate as many records as needed by variation of those parameters within their allowable range.

Due to the complex nature of the formation of seismic waves and their travel path before reaching the recording station, a stochastic approach may be most suitable for generating artificial accelerograms. In this regard, different stochastic models, both stationary and nonstationary have extensively been used in the literature to simulate earthquake ground motions. Earlier, stationary white noise random models for modeling earthquake ground motions were developed [1–5]. In this regard, the stationary filtered white noise model of Kanai–Tajimi has been the favorite model for many researchers and engineers [6–8]. These efforts were followed by introduction of the nonstationary stochastic models for generating earthquake accelerograms [9–14]. More recently,

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progress was made in improving these models to include the nonstationary nature of the accelerograms frequency contents [15–20].

In this study, an efficient procedure for modeling and generation of artificial accelerograms for any site with at least one existing earthquake record is presented. The well-known Kanai–Tajimi model is modified to include both the nonstationary natures of the amplitude and frequency content of the earthquake records. Specific examples of Naghan (1977), Tabas (1978) and Manjil (1990) earthquakes, which recently occurred in Iran, are treated. It is shown that the produced accelerograms maintain the time dependent frequency content of the original records. Comparisons have also been made between the response spectra of the simulated earthquake records with those of the original ones to show the applicability of the approach.

2. Nonstationary Kanai–Tajimi model

Based on Kanai's investigation regarding the frequency content of different earthquake records, Tajimi proposed the following relation for the spectral density function of the strong ground motion with a distinct dominant frequency:

$$G(\omega) = \frac{[1 + 4\xi_g^2(\omega/\omega_g)^2]}{[1 - (\omega/\omega_g)^2]^2 + 4\xi_g^2(\omega/\omega_g)^2} G_0 \quad (1)$$

Here γ_g and ω_g are the site dominant damping coefficient and frequency, and G_0 is the constant power spectral intensity of the bed rock excitation. In practice these parameters need to be estimated from the local earthquake records and/or site geological features. The Kanai–Tajimi power spectral density function may be interpreted as corresponding to an 'ideal white noise' excitation at the bedrock level filtered through the overlying soil deposits at a site. The most serious shortcoming of the original Kanai–Tajimi model is its treatment of earthquakes as stationary random processes. An improved version of the model was introduced in [20] to capture the nonstationary feature of the real earthquake records. This generalized nonstationary Kanai–Tajimi model is represented by the following equations:

$$\ddot{X}_f + 2\xi_g(t)\omega_g(t)\dot{X}_f + \omega_g^2(t)X_f = n(t) \quad (2)$$

$$\ddot{X}_g = -(2\xi_g(t)\omega_g(t)\dot{X}_f + \omega_g^2(t)X_f)e(t) \quad (3)$$

where X_f is the filtered response, $\omega_g(t)$ is the time dependent ground frequency, $\xi_g(t)$ is the effective ground damping coefficient, $\ddot{X}_g$ is the output ground acceleration, and $e(t)$ is the amplitude envelope function.

In Eq. (2), $n(t)$ is a stationary Gaussian white noise process with the following parameters:

$$E[n(t)] = 0 \quad E[n(t_1)n(t_2)] = 2\pi G_0 \delta(t_1 - t_2) \quad (4)$$

where $E[\cdot]$ stands for the expected value, G_0 is the constant power spectral intensity of the noise, and $\delta(\cdot)$ is the Dirac delta function. For $e(t)=1$, and $\omega_g(t)$ and $\xi_g(t)$ being constants, Eqs. (2)–(4) reduce to the original Kanai–Tajimi model. In this case the power spectral density of the ground acceleration is given by Eq. (1).

Eqs. (2)–(4) provides a filtered white noise stochastic time series with appropriate frequency content and amplitude modulation for ground acceleration during earthquakes. Therefore, the model can be conveniently used for stochastic response analysis of structures to earthquake excitations.

3. Modeling procedure

In this approach, time-varying parameters for dynamic version of the Kanai–Tajimi model are considered. In order to estimate the time-dependency of the filter parameters, at least one recorded accelerogram is needed. For this purpose the 'Moving-Time-Window' technique is used. This method is based on the assumption that a nonstationary process can approximately be assumed to be stationary within a time-window with appropriate size. Statistical methods are then used to evaluate the time-dependent amplitude envelope and the evolutionary ground frequency. The estimated parameters corresponding to a certain window temporal position is assigned to the center point of that window. This parameter estimation process is then repeated for successive window positions.

In the present study, $\xi_g(t)$ is assumed to be a constant, and the time-evolution of $\omega_g(t)$ and $e(t)$ are determined using the following steps:

3.1. Determination of the duration of strong motion

The duration of the earthquake record is obtained by eliminating the beginning and final part of the accelerogram containing less than 5% of the cumulative energy of the original record.

3.2. Estimation of the moving time-window size

The time-window should be sufficiently short to capture the rapid changes in frequency content, but long enough to provide for stable estimation of parameters and the ability to capture significant low frequency components. In this study, the optimal window size is selected based on the frequency content of the earthquake using a trial and error method. Attention was also given to the stability of the parameter estimations for each record.

3.3. Estimation of the amplitude envelope function $e(t)$

Using a time-window that moves from the beginning to the end of the accelerogram, the standard deviation of the envelope function within each time window is calculated as follows:

$$\sigma = (E[X^2] - E[X]^2)^{1/2} \quad (5)$$

This value is assigned to the center point of each time

window. Then, a smooth algebraic time function $\sigma_a(t)$ is fitted to the time variation of the standard deviation. Finally, the amplitude envelope function is defined as:

$$e(t) = C_0 \cdot \sigma_a(t) \quad (6)$$

where C_0 is a constant that is used to normalize the mean intensity of the synthetic accelerograms to the intensity of original record.

Table 1
Characteristics of actual records

Earthquake	Magnitude (M_b)	Soil condition ^a	Epicentral distance	PGA (cm/s^2)	Record duration (s)	Effective duration (s)	$\int a^2(t)dt$ (cm^2/s^3)
Naghan	5.4	Very dense (S_B – S_C)	7 (km)	709.46	20.9	5	183 531
Tabas	6.7	Stiff (S_C – S_D)	27 (km)	915.39	25	15	643 117
Manjil	6.4	Very dense (S_B – S_C)	43 (km)	517.96	53.4	30	272 479

^a Based on UBC97 soil profile types.

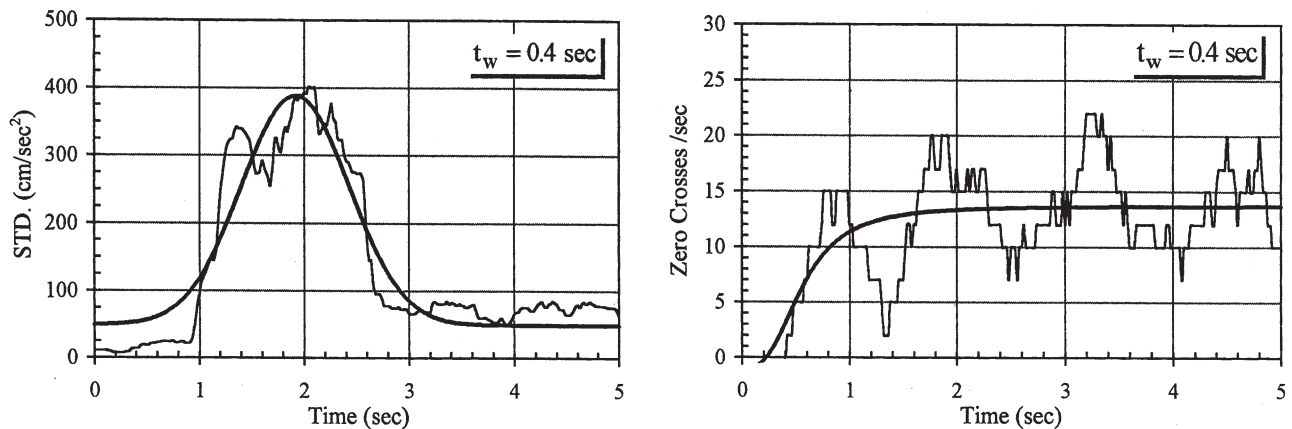


Fig. 1. Variation of standard deviation and zero-crossing with time and smooth curves for Naghan earthquake.

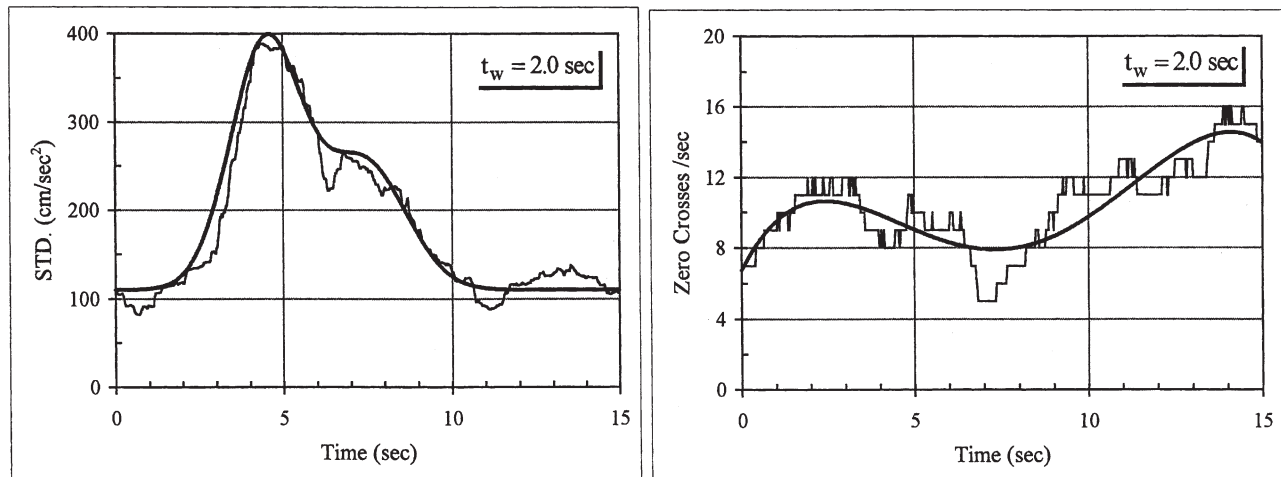


Fig. 2. Variation of standard deviation and zero-crossing with time and smooth curves for Tabas earthquake.

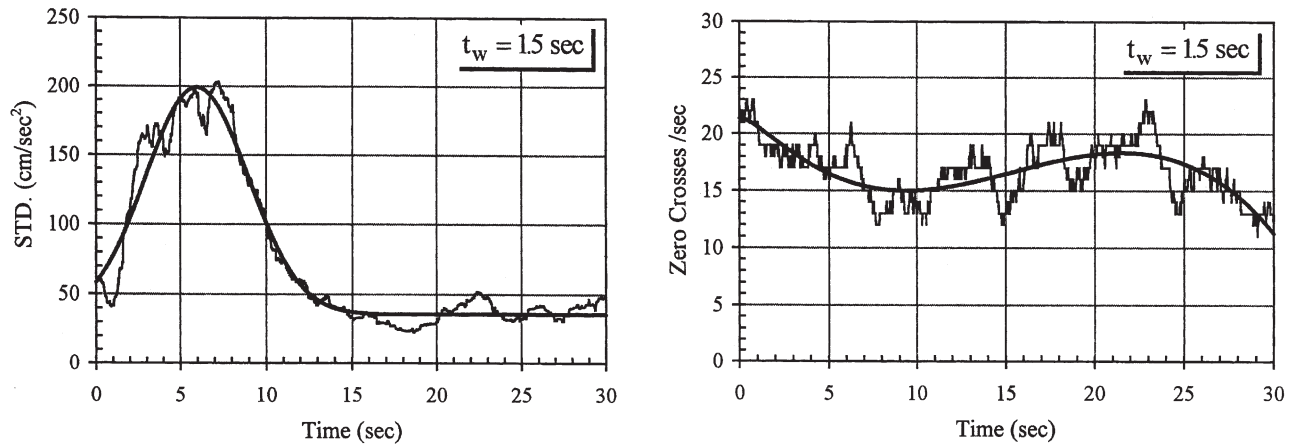


Fig. 3. Variation of standard deviation and zero-crossing with time and smooth curves for Manjil earthquake.

Table 2
Statistical of an ensemble of 50 synthetic accelerograms

Earthquake	Peak acc. (cm/s ²)		$\int a^2(t)dt$ (cm ² /s ³)	
	Mean	SD	Mean	SD
Naghan	860.7	176.1	167 354	15 233
Tabas	922.4	165.7	625 508	75 133
Manjil	568.6	88.4	274 572	21 409

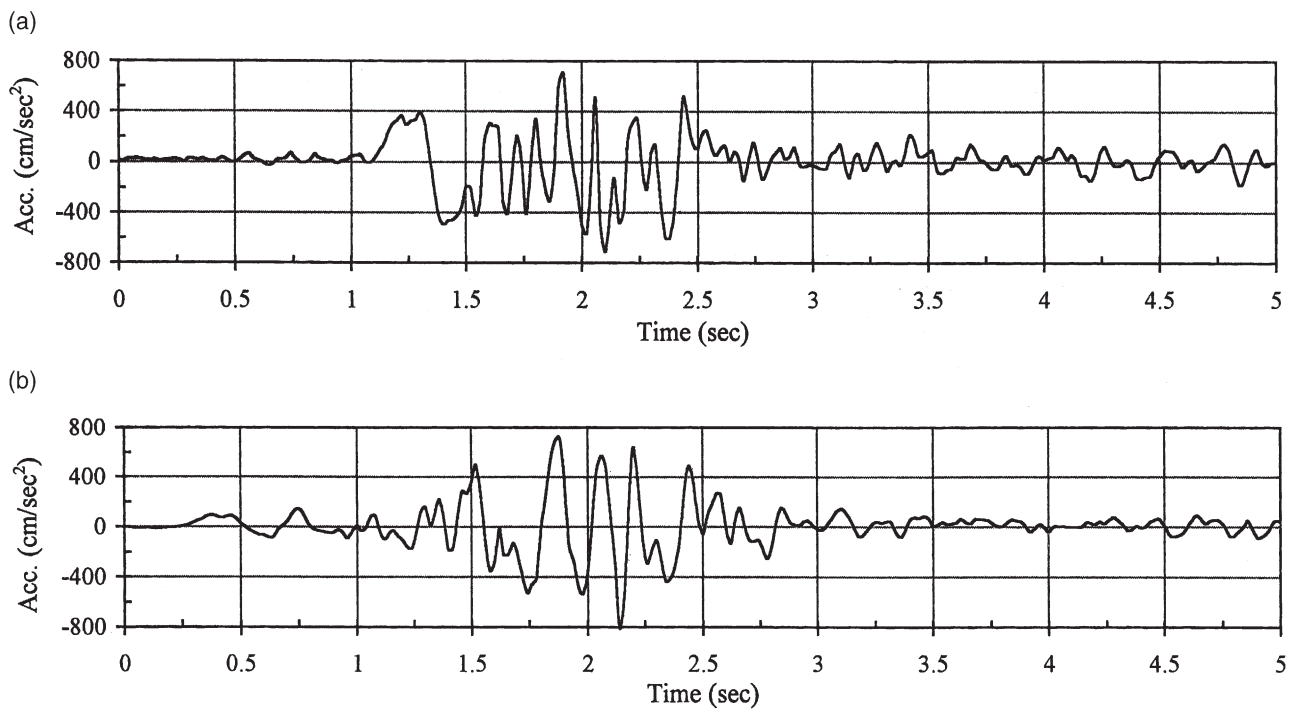


Fig. 4. (a) Actual, and (b) artificial earthquake ground acceleration of Naghan record.

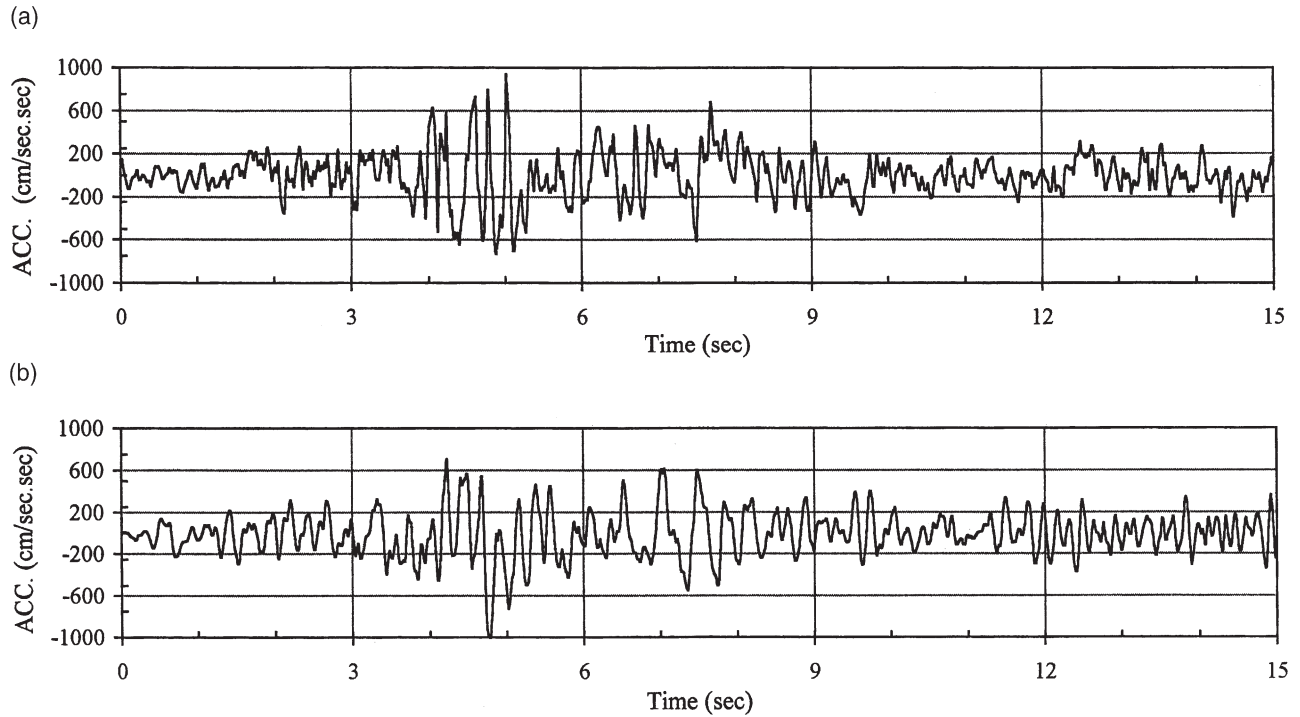


Fig. 5. (a) Actual, and (b) artificial earthquake ground acceleration of Tabas record.

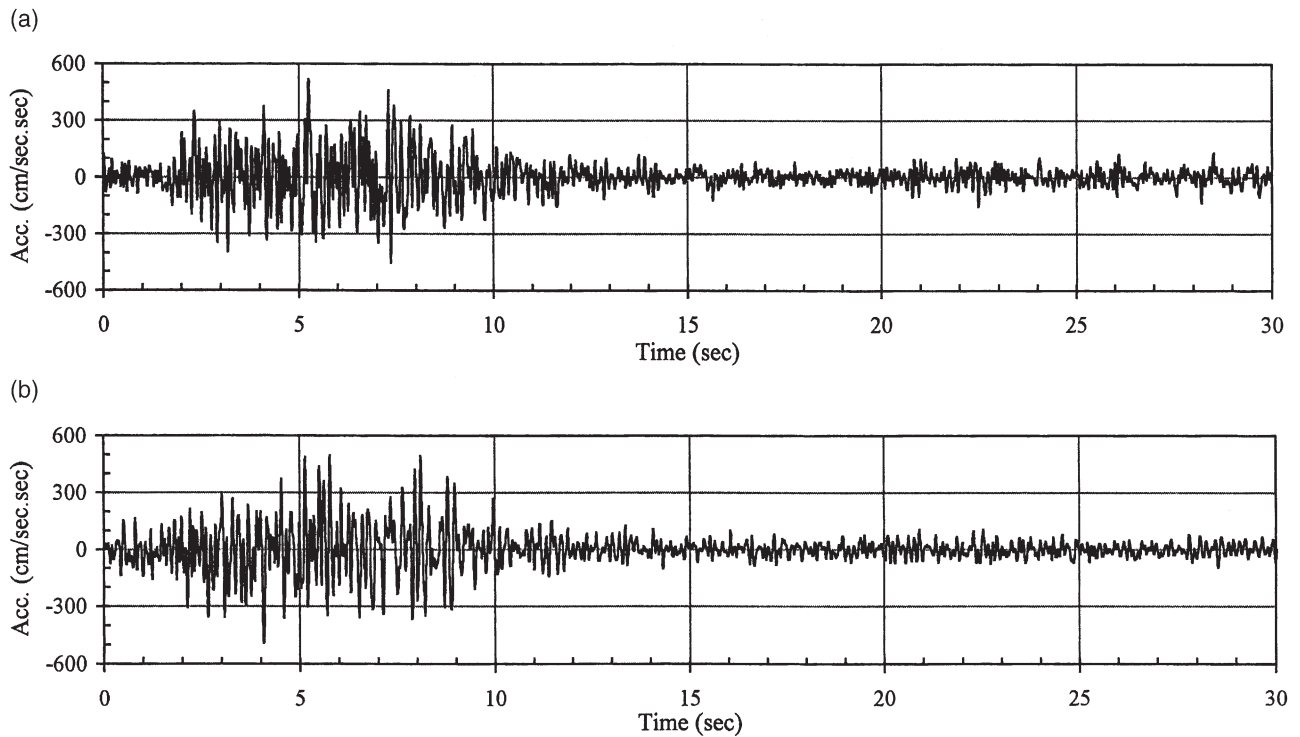


Fig. 6. (a) Actual, and (b) artificial earthquake ground acceleration of Manjil record.

3.4. Evaluation of the ground frequency function $\omega_g(t)$

A good measure of frequency content is the number of zero axis crossing per second. In order to determine the time variation of the frequency content of an earth-

quake record, the moving-time-window procedure is used. The time window size is assumed to be the same as the one used for the amplitude estimation. The zero-crossing rate at time, t , is defined as:

$$\hat{F}_C(t) = \left(\text{number of zero} \right. \\ \left. - \text{crossing within the time interval} \left(t \pm \frac{t_w}{2} \right) \right) \bigg|_{t_w} \quad (7)$$

where t_w is the time-window size. A suitable algebraic time function is then fitted to the variation of the zero-crossing rate $\hat{F}_C(t)$ with time for each accelerogram. The time-dependent ground frequency function then is defined as:

$$\omega_g(t) = \pi \hat{F}_C(t) \quad (8)$$

With the analytical expression for the amplitude and frequency, the artificial ground motion can be generated in the following manner:

1. Generation of a stationary, Gaussian, white noise stochastic time series with the appropriate intensity.

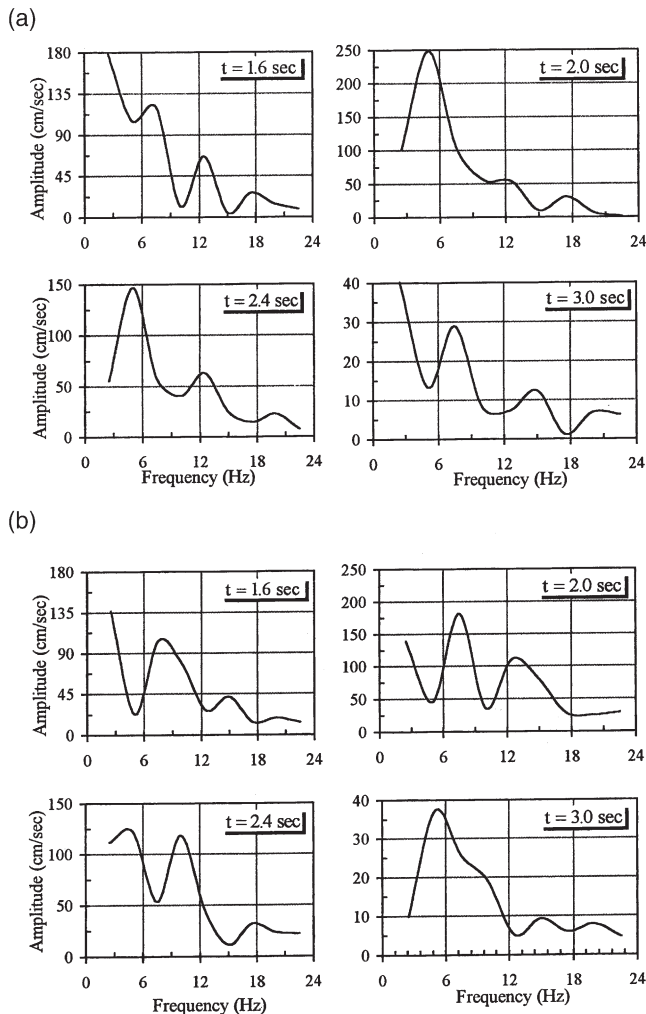


Fig. 7. Time evolution of Fourier spectra of (a) the simulated, and (b) the original for Naghan earthquake.

2. Numerical integration of Eqs. (2) and (3) with the parameters, $\xi_g(t)$, $e(t)$ and $\omega_g(t)$ determined from the previous steps for evaluating $\ddot{X}_g(t)$.

As was noted before, the procedure outline here is useful for generating synthetic earthquake time-histories for sites for which at least one earthquake record is available. For sites with no earthquake records, one needs to estimate the time evolution of the frequency and envelope functions, possibly through analysis of seismic wave propagation and/or with analogy with the geologically similar sites.

4. Applications

The generalized Kanai–Tajimi model is used to produce artificial records for three Iranian earthquake accelerograms with different characteristics. These are Naghan (1977), Tabas (1978), and Manjil (1990) earthquakes. The earthquake magnitude, peak acceleration, acceleration intensity, soil conditions, epicentral distance

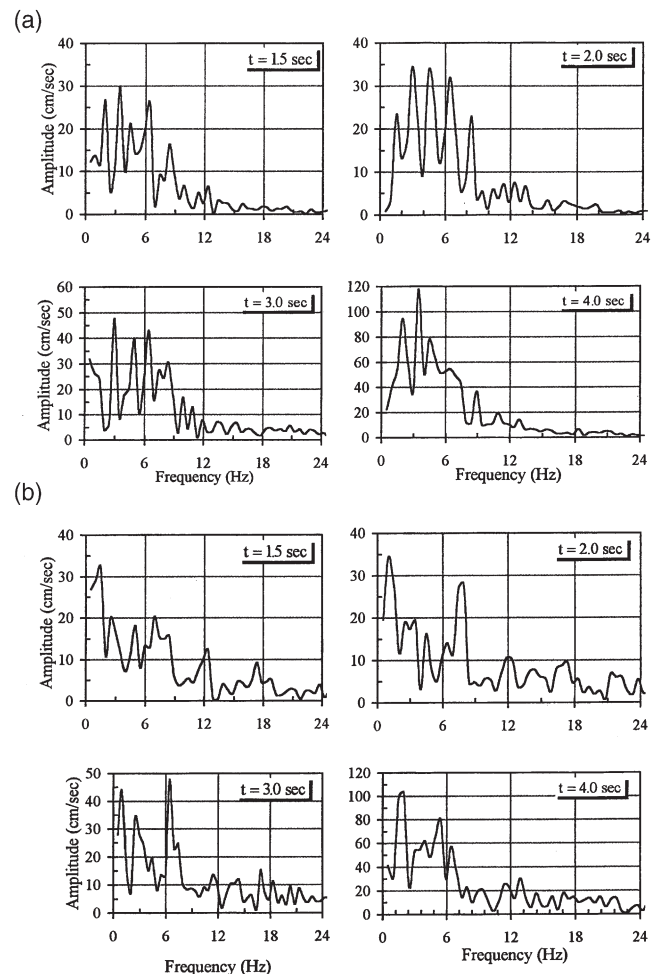


Fig. 8. Time evolution of Fourier spectra of (a) the simulated, and (b) the original for Tabas earthquake.

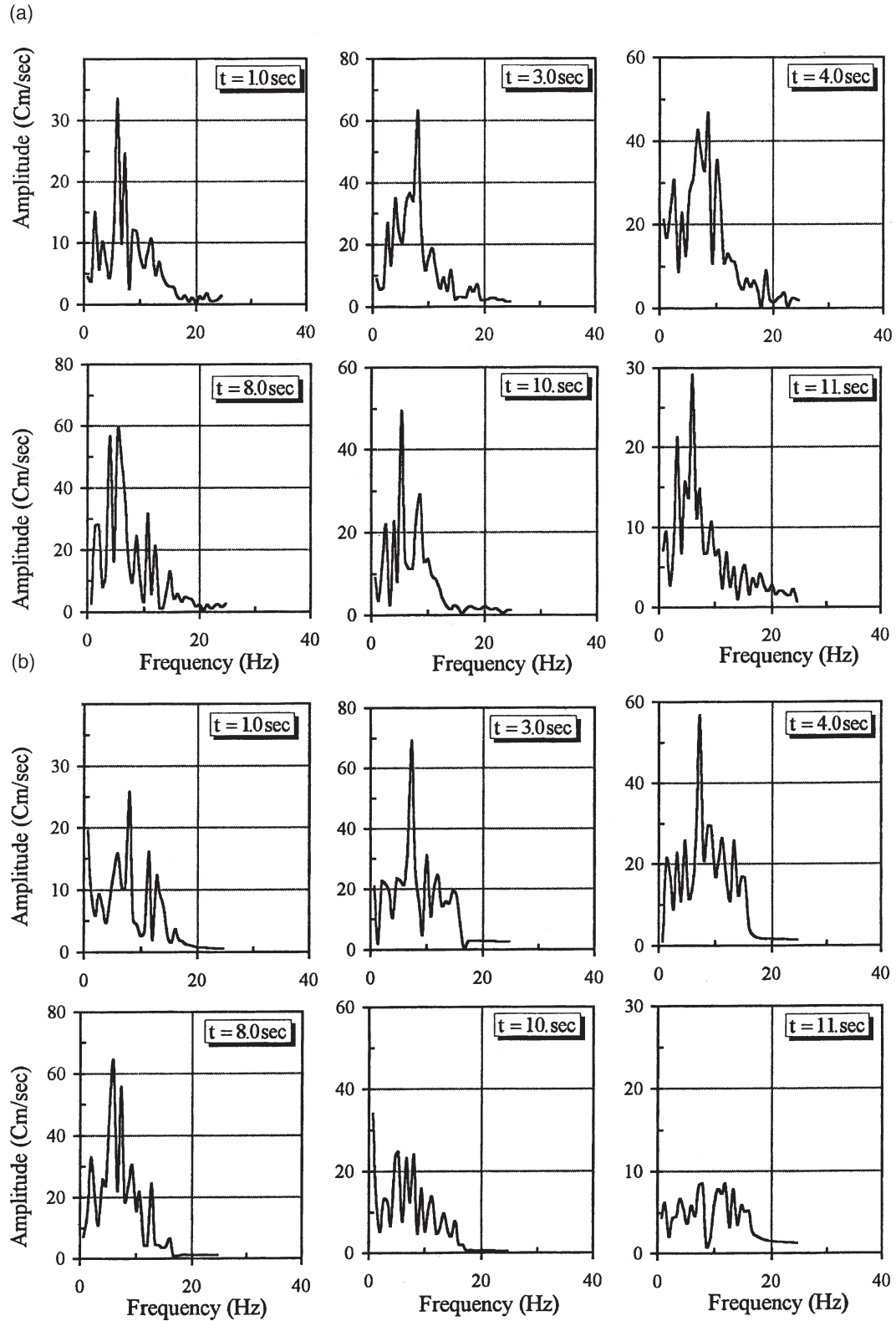


Fig. 9. Time evolution of Fourier spectra of (a) the simulated, and (b) the original for Manjil earthquake.

and duration of strong motion records are presented in Table 1. Duration of strong motion for each record is determined based on the Trifunac and Brady method.

For identifying the model parameters, window sizes 0.4, 2 and 1.5 s are used for the Naghan, Tabas and Manjil earthquakes, respectively. As noted before, the window size should be selected in such a way that it captures the time evolutions of the significant frequency content and the amplitude of the record. That is, it should sufficiently long to capture the important long period component of the excitation and also lead to stable estimation of the model parameters. The window also should be sufficiently short compare to the time scale of time evolution of the frequency and amplitude of the record.

For these records, the variation of the standard deviation and zero-crossing rate with time are shown in Figs. 1–3. In these figures, the solid lines correspond to the algebraic time functions that are fitted to the data through regression analysis. In the following, the explicit expressions for the smooth curve functions for Naghan, Tabas and Manjil are given by Eqs. (9)–(11) respectively:

Naghan

$$\sigma_a(t) = 49.1 + 339.84 \exp\left(-0.5 \left(\frac{t-1.917}{0.5134}\right)^2\right)$$

$$\hat{F}_C(t) = -0.998 + \frac{14.71}{(1+1/(t/0.558)^{2.842})} \quad (9)$$

Tabas

$$\sigma_a(t) = 145 \exp(-0.33(t-7.4)^2) + 280 \exp(-0.44(t-4.5)^2) + 110 \quad (10)$$

$$\hat{F}_C(t) = -0.00384t^4 + 0.1219t^3 - 1.1881t^2 + 3.8289t + 6.7457$$

Manjil

$$\sigma_a(t) = 35.6677 + 163.5226 \exp[-0.5((t-5.9473)/2.9983)^2]$$

$$\hat{F}_C(t) = 21.3648 + 0.5043t - 1.8943t^{1.5} + 0.7133t^2 - 0.0722t^{2.5} \quad (11)$$

Figs. 1–3 show that the smooth curve functions are in good agreement with the standard deviation and zero-crossing rate of the original accelerograms. The amplitude, standard deviation, envelope function and the ground frequency functions are calculated from Eqs. (6) and (8). Also, 0.35, 0.35, and, 0.3 are, respectively, used for the ground damping coefficient $\xi_g(t)$, of the Naghan, Tabas and Manjil earthquakes.

For producing the synthetic accelerograms, a segment of the stationary white noise process with an appropriate duration is first generated. A constant power spectral

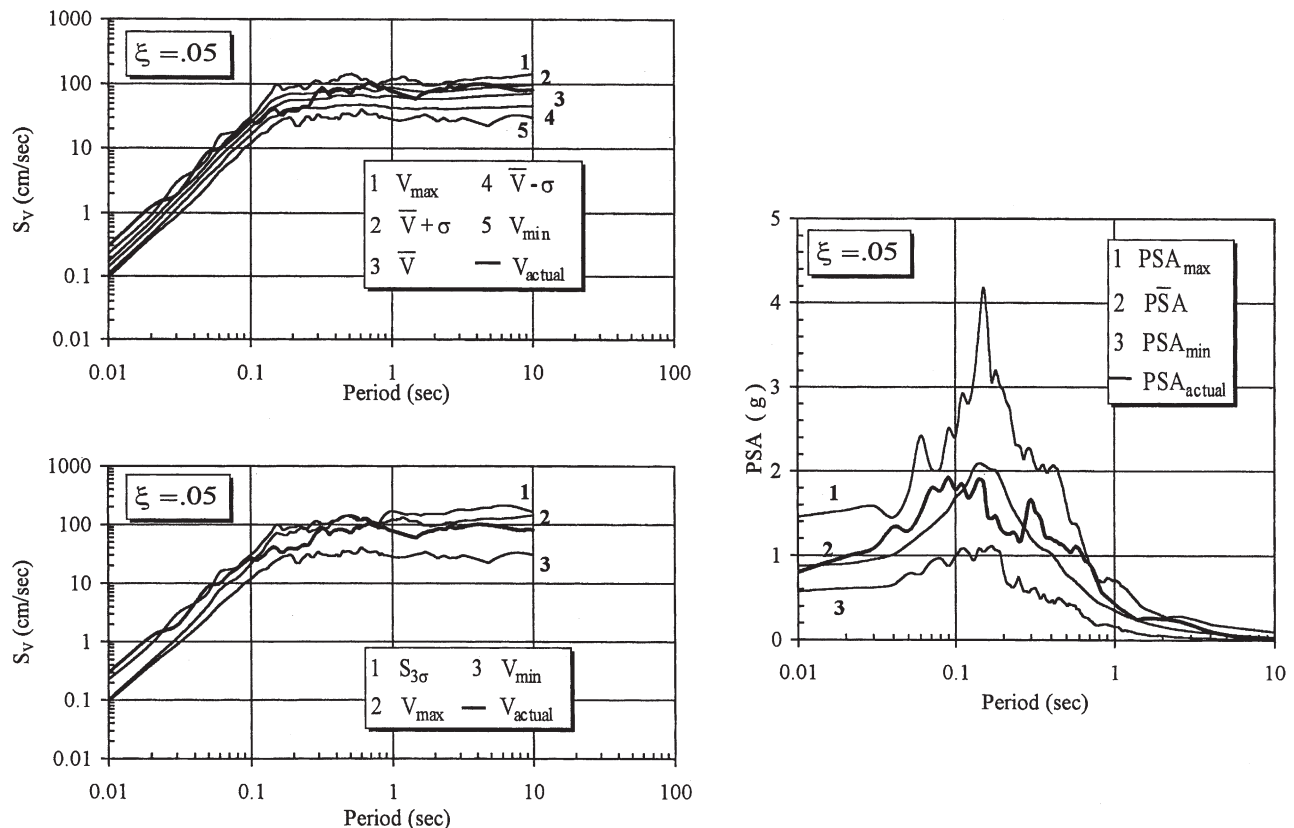


Fig. 10. Comparison between responses spectra of original and statistics of simulated accelerograms for Naghan earthquake.

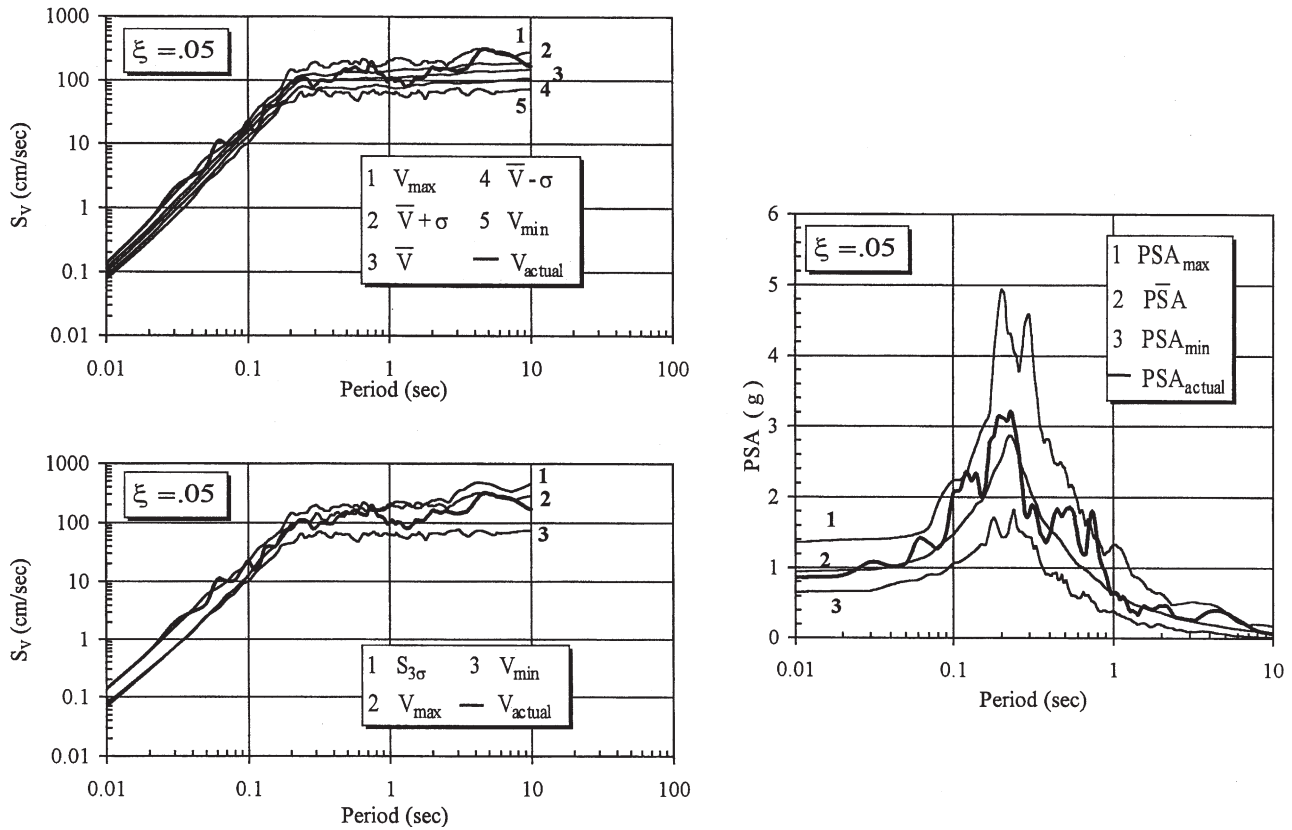


Fig. 11. Comparison between responses spectra of original and statistics of simulated accelerograms for Tabas earthquake.

intensity of $G_0=1 \text{ cm}^2/\text{s}^3$ and time interval of $\Delta T=0.02$, are used for the white noise process generation. The analytical expressions for $\omega_g(t)$ and $e(t)$, that was determined before, are used and Eq. (2) is integrated numerically for evaluating the filter response. The time history of the ground acceleration is then computed from Eq. (2). For each record ensembles of 750 samples are generated and the value of C_0 is determined such that the expected total energy of synthetic accelerograms becomes equal to that of the original record within the predetermined duration. Following this procedure, the values of $C_0=0.074$, 0.072 and 0.06 are found for the simulated Naghan, Tabas and Manjil earthquakes, respectively.

The suitability of the simulation procedures is investigated using different criteria. First, an ensemble of 50 synthetic accelerogram is generated and statically studied for each earthquake. Table 2 compares the statistical characteristics of these accelerograms with those of the actual records. Figs. 4–6 compare the accelerograms of the actual Naghan, Tabas and Manjil earthquakes with the synthetically generated records. It is shown that the simulated accelerograms preserve the time dependency of the amplitude and frequency of the original records.

Next, the time evolution of Fourier spectra of the originals and the simulated accelerograms are displayed in Figs. 7–9. As was noted before, time-windows of 0.4, 2.0 and 1.5 s are respectively used in these analyses for

Naghan, Tabas and Manjil earthquakes. The aliasing effects caused by the fast Fourier transform are assumed to be negligible. The Fourier spectra give an indication of the instantaneous frequency content over the chosen window. These figures show that the time evolutions of the frequency content of the actual and the simulated accelerograms are comparable.

Statistical response spectra of the synthetic accelerograms are also compared with those of actual records. The equation governing the response of a single degree-of-freedom (SDOF) structure under the ground acceleration, $\ddot{X}_g$, is given as:

$$\ddot{X} + 2\xi_0\omega_0\dot{X} + \omega_0^2X = -\ddot{X}_g \quad (12)$$

where ω_0 and ξ_0 are the fundamental frequency and the damping coefficient of the SDOF. The velocity response spectrum is defined as:

$$S_v(T_0, \xi_0) = \max\{\dot{X}(t)\} \quad (13)$$

where $T_0=2\pi/\omega_0$ is the natural period of the structure.

For 50 samples of synthetic accelerograms the velocity response spectra are evaluated, compiled and statistically analyzed. The mean $\bar{V}(T_0, \xi_0) = E[S_v(T_0, \xi_0)]$ and the standard deviation $\sigma_v(T_0, \xi_0) = [E\{(S_v(T_0, \xi_0) - \bar{V})^2\}]^{1/2}$ of the resulting velocity response spectra are computed. In addition, the absolute maximum velocity $V_{\max}(T_0, \xi_0) = \max\{S_v(T_0, \xi_0)\}$ and the minimum velocity,

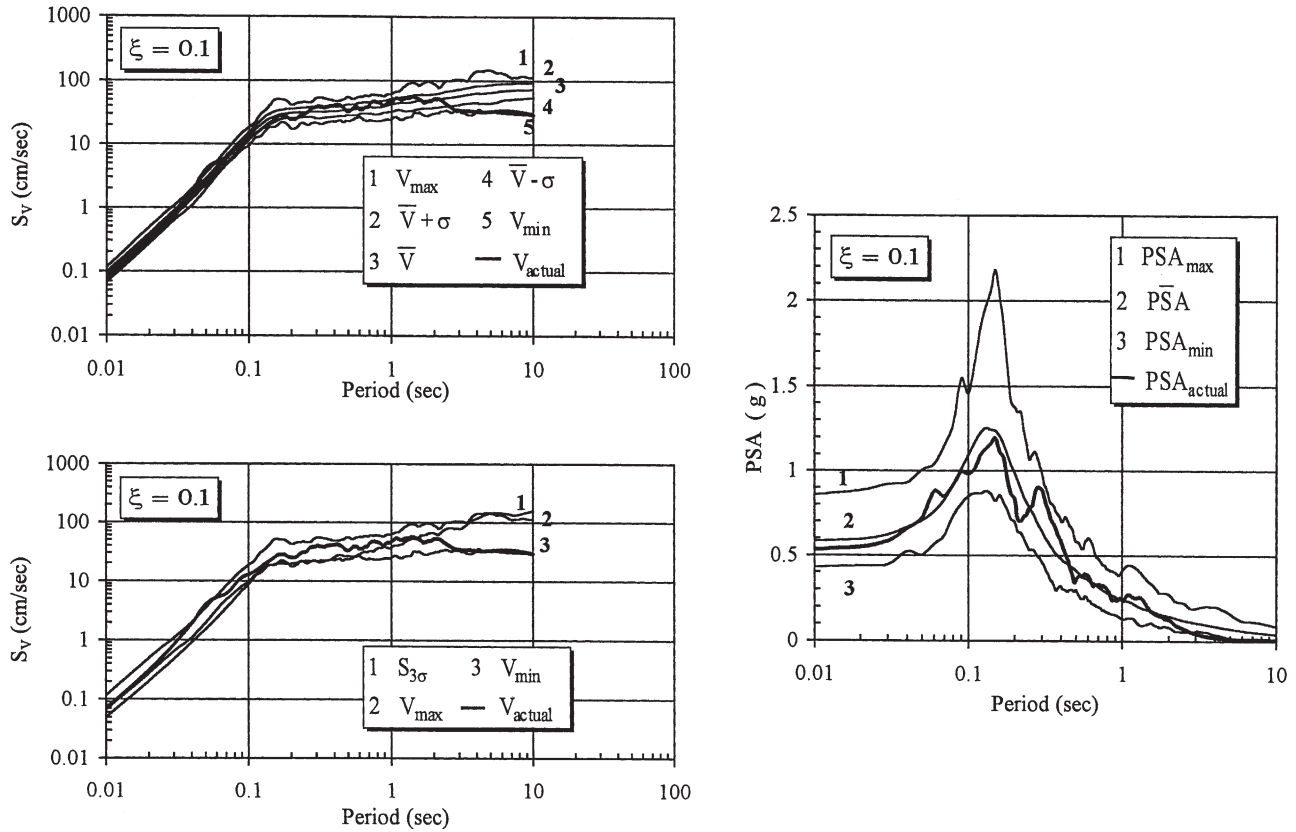


Fig. 12. Comparison between responses spectra of original and statistics of simulated accelerograms for Manjil earthquake.

$V_{\min}(T_0, \xi_0) = \min\{S_V(T_0, \xi_0)\}$ of the response spectra for all samples are evaluated. Figs. 10–12 compare the velocity response spectra and the related statistics for the ensemble of the 50 simulated accelerograms with those for the actual records for different values of damping coefficient. These statistics include the mean velocity response spectra, $\bar{V}$, the absolute maximum velocity, $V_{\max}$, the minimum velocity, $V_{\min}$, and $\bar{V} \pm \sigma_V$. These figures also compare the pseudo-acceleration response spectra for the synthetic accelerograms with those for the actual ones. It is shown that in all cases the response spectra for the synthetic accelerograms are close to those for the actual earthquakes. Also, $\bar{V} + \sigma_V$ curves appear to provide a reasonable bound on the actual velocity response spectra. For periods larger than about 5 s, however the velocity response spectra for the actual record are quite low and are in agreement with $\bar{V} - \sigma_V$ estimates of the simulated accelerograms. Figs. 10–12 also show that the $\bar{V} + \sigma_V$ may be used as a basis for developing smooth site-dependent response spectra.

For different natural periods and damping coefficients, the following spectral parameter:

$$S_V(T_0, \xi_0) = \max\{3\sigma_{\dot{X}}(t)\}, \quad \sigma_{\dot{X}}(t) = E[\dot{X}^2]^{1/2}. \quad (14)$$

are evaluated, as the results are shown in Figs. 10–12. The velocity response spectra for the actual record and the simulated response spectra $V_{\max}$ and $V_{\min}$ are also

shown in these figures for comparison. It is shown that the present three times standard deviation spectra, $S_{3\sigma}$, provide a reasonable bound on the peak velocity responses for all earthquakes.

5. Conclusions

The effectiveness of the nonstationary Kanai–Tajimi model in generating artificial accelerograms is investigated. The technique is based on a generalization of the original Kanai–Tajimi model to include the nonstationary evolution of amplitude and dominant frequency of ground motion. The time-varying model parameters for three Iranian earthquakes that occurred on stiff/and or very dense soils are estimated using actual accelerograms and a moving time-window. The model was used in generation of artificial earthquake records for The Naghan, Tabas and Manjil earthquakes. A large number of synthetic accelerograms are generated and statistically analyzed. The suitability of the model is investigated based on the comparison of the statistical characteristics of the generated accelerograms with those of the original ones. While the procedure is rather simple to use, the study shows that the model is able to capture the nonstationary nature of the amplitude and frequency content of the strong ground motions. The statistical response

spectra for various earthquakes are also evaluated and the results are favorably compared with those of actual records.

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